

Contest Math 8: Intro to Geometry (27079)

Connor Gordon

Sunday

Feb 16, 2025 - Nov 9, 2025

12:45 - 2:45 PM ET

Homework

Meeting: 1 2 3 4 5 6 **7** 8 9 10 11 12 13 14 15 16 17 18 19 20 21
 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Homework: Meeting 7

Available **Mar 30**.Due **Apr 6**.

Readings

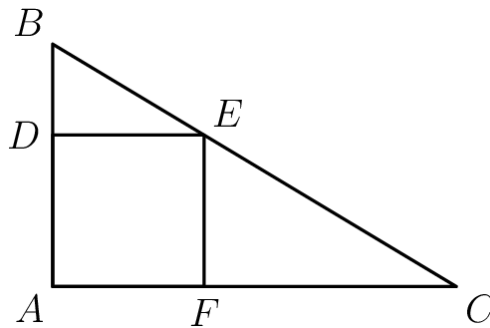
Meeting 7: Section 5.5**Meeting 8:** Sections 6.1, 6.4

Challenge Problems

Problem 1 (28912)

Problem:[Report Error](#)

In the diagram below, $ADEF$ is a square with side length 4. If $AC = 12$, what is BD ?



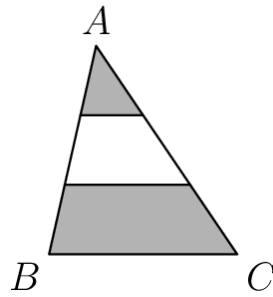
(Based on problem 5.13 in the textbook)

Your Answer

Problem 2 (23267)

Problem:[Report Error](#)

In triangle ABC , shown here, $\overline{AB}$ and $\overline{AC}$ have each been divided into three congruent segments. What fraction of triangle ABC is shaded? Express your answer as a common fraction.



If you are having trouble with this problem, see exercise 5.5.2 (b) in the textbook.

Your Answer

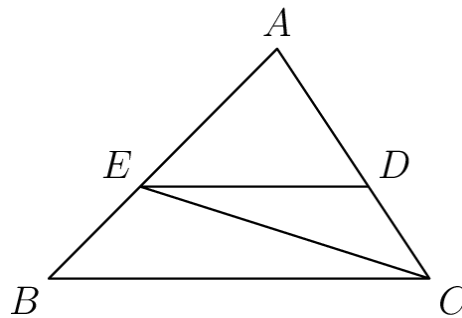
Tips

Problem 3 (23264)

Problem:

We know $\overline{DE} \parallel \overline{BC}$. Given $[ADE] = 9$ and $[CDE] = 3$, find $[ABC]$.

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If you are having trouble with this problem, see problem 5.18 in the textbook.

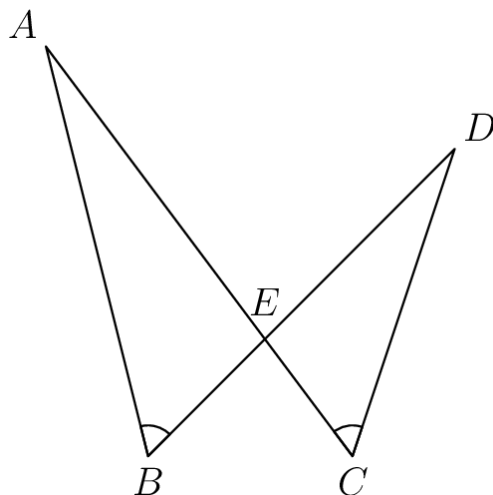
Your Answer

Problem 4 (23268)

Problem:

We know $\angle B = \angle C$. Given $AB = 50$, $BE = 20$, $DC = 35$, and $CA = 51$, find DE .

Report Error



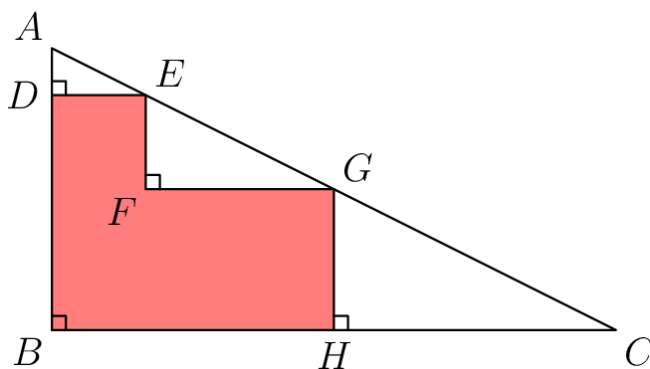
[Tips](#)

Problem 5 (39139)

Problem:

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Right triangles ABC , ADE , EFG , and GHC are all similar. Also, $[ADE] = 1$, $[EFG] = 4$, and $[GHC] = 9$. Find the area of the red region.

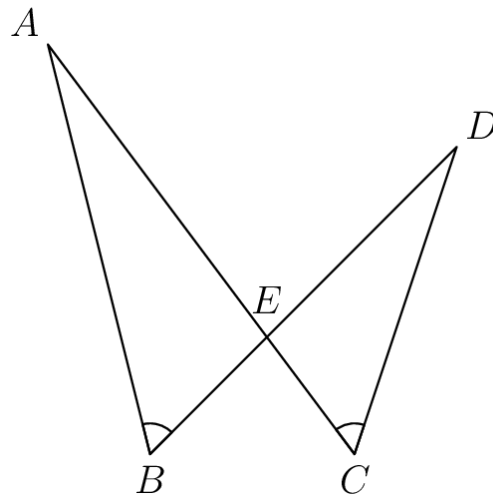


Problem 6 (23343)

Problem:

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We know $\angle B = \angle C$, $BE = 5$, and $CE = 3$. If $[AEB] = 80$, what is $[CED]$?

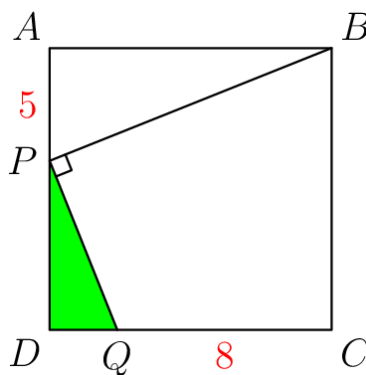


[Tips](#)

Problem 7 (39173)

Problem:

$ABCD$ is a square. Find the area of the green region.

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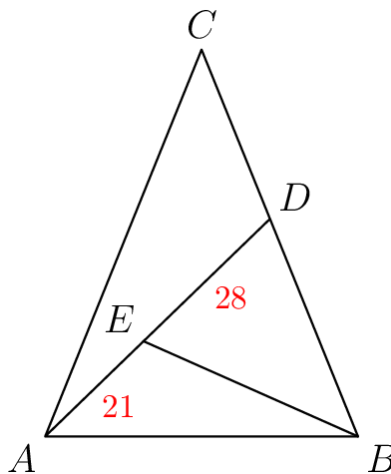
[Tips](#)

Problem 8, Part (a) (23261)

Problem:

In the diagram below, we have $\angle ABC = \angle CAB = \angle DEB = \angle BDE$, $AE = 21$, and $DE = 28$. Find BD .

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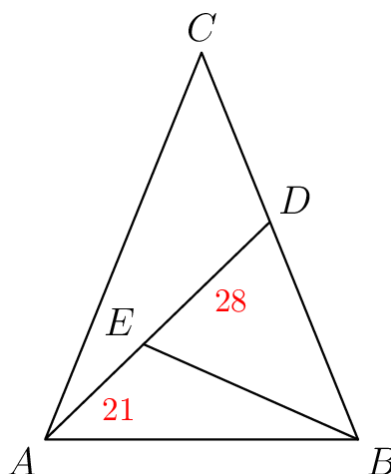


[Tips](#)

Problem 8, Part (b) (28913)

Problem:
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In the diagram below, we have $\angle ABC = \angle CAB = \angle DEB = \angle BDE$. Find CA .

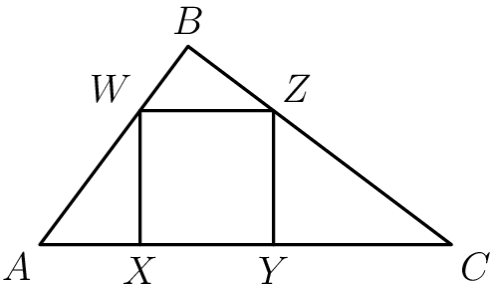


[Tips](#)

Problem 9 (23262)

Problem:
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Right triangle ABC has side lengths $AB = 1$, $BC = 2$, and $AC = \sqrt{5}$. Square $XYZW$ is inscribed in triangle ABC with X and Y on $\overline{AC}$, W on $\overline{AB}$, and Z on $\overline{BC}$. What is the area of the square?



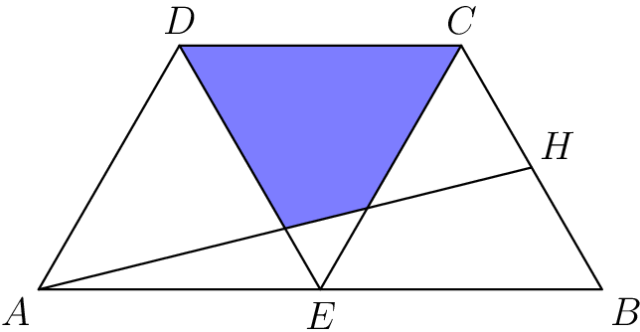
Your Answer

[Tips](#)

Problem 10 (39114)

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Problem:
Triangles ADE , CDE , and BCE are all equilateral triangles with area 1, and H is the midpoint of $\overline{BC}$. Find the area of the blue region.



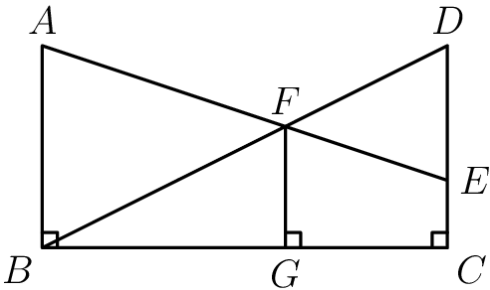
Your Answer

[Tips](#)

Problem 11 (23260)

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Problem:
In the diagram below, we have $DE = 3 \cdot EC$ and $AB = DC = 20$. Find the length FG .



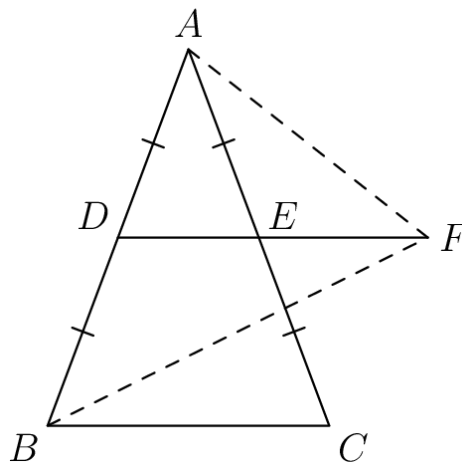
Your Answer

[Tips](#)

Problem 12 (39080)

Problem:[Report Error](#)

In the diagram, $\triangle ABC \cong \triangle BFA$. Find $\frac{AB}{BC}$.



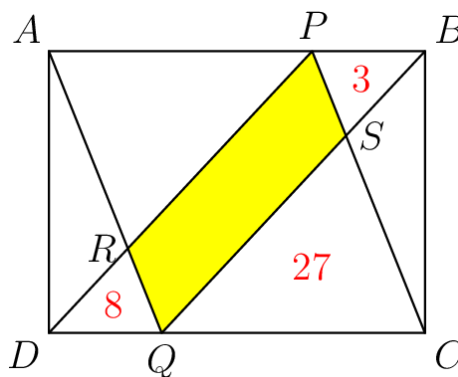
Your Answer

[Tips](#)

Problem 13 (39171)

Problem:[Report Error](#)

Rectangle $ABCD$ is divided into several regions, some of which are labelled with their areas. Find the area of the yellow region.

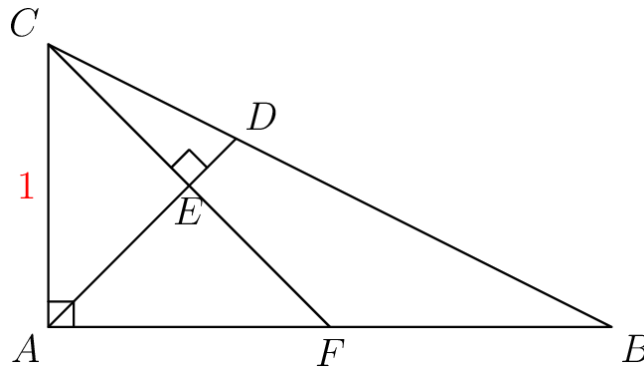


Your Answer

Problem 14 (39135)

Problem:[Report Error](#)

In the diagram, $BD = 2 \cdot CD$ and $CF = FB$. Find the length AE .

**Beast Problem:** 🐾

This is a **Beast** problem. Answering this problem can only increase your score. You will not lose points for skipping this problem or answering incorrectly.

[Tips](#)

Book Problems (39198)

In addition to the homework listed here each week, you also need to complete several additional homework problems from your textbooks. This week's book problems are:

- *Introduction to Geometry*: 5.31, 5.32, 5.35, 5.45, 5.48

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